

KITABU QUEST

P6

MATHEMATICS



Cover scene

learners using counters and measuring tools with a friendly mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

6

Title Page

KITABU QUEST P6 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P6 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Reading, writing and comparing whole numbers beyond 1
2. Multiplication and division of integers
3. Powers, indices, LCM and GCF
4. Operations on fractions
5. Rounding and conversion of decimals fractions/numbers
6. Percentages and ratios
7. Relationship between volume, capacity and mass
8. Speed, distance and time
9. Simple interest and problems involving saving
10. Equivalent expressions and number sequences
11. Regular polygons and bearings
12. Construction of polygons and nets for cuboids and prisms
13. Area bounded by a circle, surface area of cuboids and volume of a cylinder
14. Data handling

As you finish each topic, return to the success criteria and correct one answer before moving on.

Reading, writing and comparing whole numbers beyond 1: Lesson Opener

Learning focus: Reading, writing and comparing whole numbers beyond 1

Country link: a Rwandan school, market, farming, building, transport, or data problem for Reading, writing and comparing whole numbers beyond 1.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- read, write and compare whole numbers beyond 1 000 000
- identify the place beyond 1,000,000
- explain the concept of adding a 7-digit number by a 7-digit number which involves carrying
- explain the concept of subtracting larger numbers with borrowing
- explain the concept of multiplying larger numbers
- explain the concept of dividing a 7-digit

Inquiry questions:

- How can reading, writing and comparing whole numbers beyond 1 000 000 help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing and comparing whole numbers beyond 1: Learn

Reading, writing and comparing whole numbers beyond 1 teaches you to read, write, compare, and calculate with large whole numbers.

Core idea:

- A digit has value because of its position: ones, tens, hundreds, thousands, ten-thousands, and hundred-thousands.
- Compare numbers from left to right, starting with the greatest place value.
- For addition and subtraction, align place values before calculating.
- For multiplication or division, check whether the answer should be bigger or smaller than the starting number.

Rwanda example: class enrolment, harvest records, market stock, and school budget totals can all use whole-number operations.

Key words:

- Reading, writing and comparing whole numbers beyond 1
- Reading
- writing
- comparing
- whole
- numbers
- beyond
- method

Reading, writing and comparing whole numbers beyond 1: Worked Example

Worked example for Reading, writing and comparing whole numbers beyond 1: A school library has 23,475 storybooks and 18,320 revision books. How many books are there altogether?

1. Align place values.
2. Add ones, tens, hundreds, thousands, and ten-thousands.
3. $23,475 + 18,320 = 41,795$.
4. Estimate: $23,000 + 18,000 = 41,000$, so 41,795 is reasonable.

Answer: 41,795 books.

Reading, writing and comparing whole numbers beyond 1: Vocabulary And Study Skill

Use these words and study moves before you practise Reading, writing and comparing whole numbers beyond 1. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Reading, writing and comparing whole numbers beyond 1
- Reading
- writing
- comparing
- whole
- numbers
- beyond

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Reading, writing and comparing whole numbers beyond 1 without a clear example, method, or evidence.

Reading, writing and comparing whole numbers beyond 1: Guided Activity

Guided activity for Reading, writing and comparing whole numbers beyond 1:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Reading, writing and comparing whole numbers beyond 1: Practice And Correction

Practice for Reading, writing and comparing whole numbers beyond 1:

Main outcome: read, write and compare whole numbers beyond 1 000 000.

1. Solve one reading, writing and comparing whole numbers beyond 1 question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to reading, writing and comparing whole numbers beyond 1 and solve it.

Reading, writing and comparing whole numbers beyond 1: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Reading, writing and comparing whole numbers beyond 1.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing and comparing whole numbers beyond 1: Outcome Check

Outcome check for Reading, writing and comparing whole numbers beyond 1: Read, write and compare whole numbers beyond 1 000 000.

1. Add 34,785 and 21,406.
2. Estimate the answer to check it.
3. Explain one place-value step.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing and comparing whole numbers beyond 1: Outcome Check

Outcome check for Reading, writing and comparing whole numbers beyond 1: Identify the place beyond 1,000,000.

1. Add 34,785 and 21,406.
2. Estimate the answer to check it.
3. Explain one place-value step.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of integers: Lesson Opener

Learning focus: Multiplication and division of integers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Multiplication and division of integers.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- multiply and divide integers
- describe the steps taken when multiplying and dividing integers
- show and explain the concept of integers to solve problems
- apply the concepts of multiplication and division to solve problems involving integers
- carry out multiplication and division of integers
- explain how integers change in multiplication and division

Inquiry questions:

- How can multiplication and division of integers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of integers: Learn

Multiplication and division of integers teaches numbers below, at, and above zero.

Core idea:

- Positive numbers are greater than zero.
- Negative numbers are less than zero.
- Zero is the turning point.
- A number line helps you compare and order integers.

Rwanda example: temperature changes, money owed, points gained or lost in a game, and height above or below a reference point can use integers.

Key words:

- Multiplication and division of integers
- Multiplication
- division
- integers
- method
- model
- unit
- answer

Multiplication and division of integers: Worked Example

Worked example for Multiplication and division of integers: The morning temperature is 3 degrees above zero. At night it drops by 7 degrees. What is the night temperature?

1. Start at +3 on a number line.
2. Move 7 steps left because the temperature drops.
3. $+3 - 7 = -4$.

Answer: 4 degrees below zero.

Multiplication and division of integers: Vocabulary And Study Skill

Use these words and study moves before you practise Multiplication and division of integers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Multiplication and division of integers
- Multiplication
- division
- integers
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Multiplication and division of integers without a clear example, method, or evidence.

Multiplication and division of integers: Guided Activity

Guided activity for Multiplication and division of integers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Multiplication and division of integers: Practice And Correction

Practice for Multiplication and division of integers:

Main outcome: multiply and divide integers.

1. Solve one multiplication and division of integers question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to multiplication and division of integers and solve it.

Multiplication and division of integers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Multiplication and division of integers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of integers: Outcome Check

Outcome check for Multiplication and division of integers: Multiply and divide integers.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of integers: Outcome Check

Outcome check for Multiplication and division of integers: Describe the steps taken when multiplying and dividing integers.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Powers, indices, LCM and GCF: Lesson Opener

Learning focus: Powers, indices, LCM and GCF

Country link: a Rwandan school, market, farming, building, transport, or data problem for Powers, indices, LCM and GCF.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use powers and indices, and apply the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) when solving problems
- state and explain the law of multiplication and division of indices
- explain the terms "base" and "exponent" or "power"
- identify the ways of working out problems involving the LCM in a practical context
- apply the law of indices in multiplication and division
- calculate the LCM and the GCF of numbers

Inquiry questions:

- How can powers, indices, lcm and gcf help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Powers, indices, LCM and GCF: Learn

Powers, indices, LCM and GCF helps you break numbers into useful groups.

Core idea:

- A factor divides a number exactly.
- A multiple is made by multiplying a number by 1, 2, 3, and so on.
- Prime numbers have exactly two factors: 1 and the number itself.
- Classification is easier when you test divisibility carefully.

Key words:

- Powers, indices, LCM and GCF
- Powers
- indices
- method
- model
- unit
- answer
- check

Powers, indices, LCM and GCF: Worked Example

Worked example for Powers, indices, LCM and GCF: List all factors of 24.

1. Test factor pairs: 1×24 , 2×12 , 3×8 , 4×6 .
2. No whole-number factor pair is missing between 4 and 6.
3. Write the factors in order.

Answer: 1, 2, 3, 4, 6, 8, 12, 24.

Powers, indices, LCM and GCF: Vocabulary And Study Skill

Use these words and study moves before you practise Powers, indices, LCM and GCF. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Powers, indices, LCM and GCF
- Powers
- indices
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Powers, indices, LCM and GCF without a clear example, method, or evidence.

Powers, indices, LCM and GCF: Guided Activity

Guided activity for Powers, indices, LCM and GCF:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Powers, indices, LCM and GCF: Practice And Correction

Practice for Powers, indices, LCM and GCF:

Main outcome: use powers and indices, and apply the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) when solving problems.

1. Solve one powers, indices, lcm and gcf question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to powers, indices, lcm and gcf and solve it.

Powers, indices, LCM and GCF: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Powers, indices, LCM and GCF.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Powers, indices, LCM and GCF: Outcome Check

Outcome check for Powers, indices, LCM and GCF: Use powers and indices, and apply the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) when solving problems.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Powers, indices, LCM and GCF: Outcome Check

Outcome check for Powers, indices, LCM and GCF: State and explain the law of multiplication and division of indices.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Operations on fractions: Lesson Opener

Learning focus: Operations on fractions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Operations on fractions.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- apply fractions in daily life situations and solve related problems
- explain how to multiply and divide fractions
- describe how to solve problems involving fractions
- multiply and divide fractions
- solve word problems involving fractions
- appreciate the importance of fractions in daily life situations

Inquiry questions:

- How can operations on fractions help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Operations on fractions: Learn

Operations on fractions teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Operations on fractions
- Operations
- fractions
- method
- model
- unit
- answer
- check

Operations on fractions: Worked Example

Worked example for Operations on fractions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 \div 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Operations on fractions: Vocabulary And Study Skill

Use these words and study moves before you practise Operations on fractions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Operations on fractions
- Operations
- fractions
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Operations on fractions without a clear example, method, or evidence.

Operations on fractions: Guided Activity

Guided activity for Operations on fractions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Operations on fractions: Practice And Correction

Practice for Operations on fractions:

Main outcome: apply fractions in daily life situations and solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to operations on fractions and solve it.

Operations on fractions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Operations on fractions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Operations on fractions: Outcome Check

Outcome check for Operations on fractions: Apply fractions in daily life situations and solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Operations on fractions: Outcome Check

Outcome check for Operations on fractions: Explain how to multiply and divide fractions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Rounding and conversion of decimals fractions/numbers: Lesson Opener

Learning focus: Rounding and conversion of decimals fractions/numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Rounding and conversion of decimals fractions/numbers.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- round off decimals accurately, and demonstrate understanding by matching equivalent fractions and decimals
- explain how to round off decimal fractions
- describe the various steps taken when rounding off numbers
- illustrate and explain how to match fractions and decimals
- convert fractions to decimals and vice versa
- apply the to match fractions with decimals

Inquiry questions:

- How can rounding and conversion of decimals fractions/numbers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Rounding and conversion of decimals fractions/numbers: Learn

Rounding and conversion of decimals fractions/numbers teaches another way to write parts of a whole using place value.

Core idea:

- The first digit after the decimal point shows tenths.
- The second digit shows hundredths.
- Decimal numbers are compared from left to right, just like whole numbers.
- Money and measurements often use decimals.

Key words:

- Rounding and conversion of decimals fractions/numbers
- Rounding
- conversion
- decimals
- fractions/numbers
- method
- model
- unit

Rounding and conversion of decimals fractions/numbers: Worked Example

Worked example for Rounding and conversion of decimals fractions/numbers: Compare 4.35 and 4.5.

1. Write 4.5 as 4.50 so both numbers show hundredths.
2. Compare whole numbers: both are 4.
3. Compare tenths: 3 tenths is less than 5 tenths.

Answer: 4.35 is less than 4.5.

Rounding and conversion of decimals fractions/numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Rounding and conversion of decimals fractions/numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Rounding and conversion of decimals fractions/numbers
- Rounding
- conversion
- decimals
- fractions/numbers
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Rounding and conversion of decimals fractions/numbers without a clear example, method, or evidence.

Rounding and conversion of decimals fractions/numbers: Guided Activity

Guided activity for Rounding and conversion of decimals fractions/numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Rounding and conversion of decimals fractions/numbers: Practice And Correction

Practice for Rounding and conversion of decimals fractions/numbers:

Main outcome: round off decimals accurately, and demonstrate understanding by matching equivalent fractions and decimals.

1. Write 3 tenths as a decimal.
2. Order 2.05, 2.5, and 2.15 from smallest to largest.
3. Use a money or measurement example to compare two decimal numbers.

Answer check:

Answers: 3 tenths = 0.3. Order: 2.05, 2.15, 2.5.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to rounding and conversion of decimals fractions/numbers and solve it.

Rounding and conversion of decimals fractions/numbers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Rounding and conversion of decimals fractions/numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Rounding and conversion of decimals fractions/numbers: Outcome Check

Outcome check for Rounding and conversion of decimals fractions/numbers: Round off decimals accurately, and demonstrate understanding by matching equivalent fractions and decimals.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Rounding and conversion of decimals fractions/numbers: Outcome Check

Outcome check for Rounding and conversion of decimals fractions/numbers: Explain how to round off decimal fractions.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentages and ratios: Lesson Opener

Learning focus: Percentages and ratios

Country link: a Rwandan school, market, farming, building, transport, or data problem for Percentages and ratios.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- work out percentages, and ratios and apply this knowledge to solve real life mathematical problems
- convert percentages to decimals and vice versa
- share quantities in a ratio
- apply percentages and ratios in solving mathematical problems
- acknowledge the importance of percentages and ratios in daily life situations
- respect one another when working in groups to solve problems involving percentages and ratios

Inquiry questions:

- How can percentages and ratios help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentages and ratios: Learn

Percentages and ratios teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Percentages and ratios
- Percentages
- ratios
- method
- model
- unit
- answer
- check

Percentages and ratios: Worked Example

Worked example for Percentages and ratios: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Percentages and ratios: Vocabulary And Study Skill

Use these words and study moves before you practise Percentages and ratios. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Percentages and ratios
- Percentages
- ratios
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Percentages and ratios without a clear example, method, or evidence.

Percentages and ratios: Guided Activity

Guided activity for Percentages and ratios:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Percentages and ratios: Practice And Correction

Practice for Percentages and ratios:

Main outcome: work out percentages, and ratios and apply this knowledge to solve real life mathematical problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to percentages and ratios and solve it.

Percentages and ratios: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Percentages and ratios.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Percentages and ratios: Outcome Check

Outcome check for Percentages and ratios: Work out percentages, and ratios and apply this knowledge to solve real life mathematical problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentages and ratios: Outcome Check

Outcome check for Percentages and ratios: Convert percentages to decimals and vice versa.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Relationship between volume, capacity and mass: Lesson Opener

Learning focus: Relationship between volume, capacity and mass

Country link: a Rwandan school, market, farming, building, transport, or data problem for Relationship between volume, capacity and mass.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- convert accurately between units of volume, capacity and mass in real-life and mathematical contexts
- state the units of length, capacity, and mass when solving problems
- explain the relationship between volume, capacity, and mass in the case of water
- convert between units of volume, capacity, and mass
- solve problems involving the relationship among volume, capacity, and mass measurements
- show respect for one another when working in groups

Inquiry questions:

- How can relationship between volume, capacity and mass help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Relationship between volume, capacity and mass: Learn

Relationship between volume, capacity and mass teaches how heavy objects are.

Core idea:

- Mass is measured with units such as grams and kilograms.
- A balance or scale helps compare mass fairly.
- Convert units before adding or comparing if the units are different.

Key words:

- Relationship between volume, capacity and mass
- Relationship
- between
- volume
- capacity
- mass
- method
- model

Relationship between volume, capacity and mass: Worked Example

Worked example for Relationship between volume, capacity and mass: A bag of beans has a mass of 3 kg. A small packet has a mass of 500 g. What is the total mass in grams?

1. Convert 3 kg to grams: $3 \text{ kg} = 3000 \text{ g}$.
2. Add $3000 \text{ g} + 500 \text{ g} = 3500 \text{ g}$.

Answer: 3500 g.

Relationship between volume, capacity and mass: Vocabulary And Study Skill

Use these words and study moves before you practise Relationship between volume, capacity and mass. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Relationship between volume, capacity and mass
- Relationship
- between
- volume
- capacity
- mass
- method
- model

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Relationship between volume, capacity and mass without a clear example, method, or evidence.

Relationship between volume, capacity and mass: Guided Activity

Guided activity for Relationship between volume, capacity and mass:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Relationship between volume, capacity and mass: Practice And Correction

Practice for Relationship between volume, capacity and mass:

Main outcome: convert accurately between units of volume, capacity and mass in real-life and mathematical contexts.

1. Name two objects measured in grams and two measured in kilograms.
2. Convert 2 kg to grams.
3. Compare 1500 g and 2 kg. Which is heavier?

Answer check:

Answer: $2\text{ kg} = 2000\text{ g}$, so 2 kg is heavier than 1500 g.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to relationship between volume, capacity and mass and solve it.

Relationship between volume, capacity and mass: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Relationship between volume, capacity and mass.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Relationship between volume, capacity and mass: Outcome Check

Outcome check for Relationship between volume, capacity and mass: Convert accurately between units of volume, capacity and mass in real-life and mathematical contexts.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Relationship between volume, capacity and mass: Outcome Check

Outcome check for Relationship between volume, capacity and mass: State the units of length, capacity, and mass when solving problems.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Speed, distance and time: Lesson Opener

Learning focus: Speed, distance and time

Country link: a Rwandan school, market, farming, building, transport, or data problem for Speed, distance and time.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- calculate speed, distance and time, solve problems that relate to different time zones and convert speed from km/h to m/s and vice versa
- explain the relationship between a 12-hour clock and a 24-hour clock
- define speed, distance, and time
- identify different units of speed, distance, and time
- explain what determines time zones
- tell the time using a 12-hour clock and a 24-hour clock

Inquiry questions:

- How can speed, distance and time help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Speed, distance and time: Learn

Speed, distance and time teaches how to read, write, convert, and use time.

Core idea:

- Time can be shown on analogue clocks, digital clocks, calendars, and timetables.
- 60 seconds make 1 minute, and 60 minutes make 1 hour.
- Timetables help you find start time, end time, and duration.

Key words:

- Speed, distance and time
- Speed
- distance
- time
- method
- model
- unit
- answer

Speed, distance and time: Worked Example

Worked example for Speed, distance and time: A lesson starts at 8:10 and ends at 8:50. How long is the lesson?

1. Count from 8:10 to 8:50.
2. 10 minutes to 8:20, 20 more to 8:40, 10 more to 8:50.
3. Total time = 40 minutes.

Answer: 40 minutes.

Speed, distance and time: Vocabulary And Study Skill

Use these words and study moves before you practise Speed, distance and time. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Speed, distance and time
- Speed
- distance
- time
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Speed, distance and time without a clear example, method, or evidence.

Speed, distance and time: Guided Activity

Guided activity for Speed, distance and time:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Speed, distance and time: Practice And Correction

Practice for Speed, distance and time:

Main outcome: calculate speed, distance and time, solve problems that relate to different time zones and convert speed from km/h to m/s and vice versa.

1. Read a clock showing 7:30 and write the time in words.
2. Convert 2 hours to minutes.
3. A trip starts at 9:15 and ends at 10:00. Find the duration.

Answer check:

Answers: 2 hours = 120 minutes. 9:15 to 10:00 is 45 minutes.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to speed, distance and time and solve it.

Speed, distance and time: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Speed, distance and time.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Speed, distance and time: Outcome Check

Outcome check for Speed, distance and time: Calculate speed, distance and time, solve problems that relate to different time zones and convert speed from km/h to m/s and vice versa.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Speed, distance and time: Outcome Check

Outcome check for Speed, distance and time: Explain the relationship between a 12-hour clock and a 24-hour clock.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simple interest and problems involving saving: Lesson Opener

Learning focus: Simple interest and problems involving saving

Country link: a Rwandan school, market, farming, building, transport, or data problem for Simple interest and problems involving saving.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- work out simple interest and solve problems involving saving
- define different terms such as simple interest, rate, principle, and time
- explain the importance of saving money
- solve problems involving savings
- solve problems involving the calculation of simple interest
- show confidence when calculating simple interest to be paid by the Saving and Credit Association

Inquiry questions:

- How can simple interest and problems involving saving help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simple interest and problems involving saving: Learn

Simple interest and problems involving saving teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Simple interest and problems involving saving
- Simple
- interest
- problems
- involving
- saving
- method
- model

Simple interest and problems involving saving: Worked Example

Worked example for Simple interest and problems involving saving: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 \div 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Simple interest and problems involving saving: Vocabulary And Study Skill

Use these words and study moves before you practise Simple interest and problems involving saving. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Simple interest and problems involving saving
- Simple
- interest
- problems
- involving
- saving
- method
- model

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Simple interest and problems involving saving without a clear example, method, or evidence.

Simple interest and problems involving saving: Guided Activity

Guided activity for Simple interest and problems involving saving:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Simple interest and problems involving saving: Practice And Correction

Practice for Simple interest and problems involving saving:

Main outcome: work out simple interest and solve problems involving saving.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to simple interest and problems involving saving and solve it.

Simple interest and problems involving saving: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Simple interest and problems involving saving.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Simple interest and problems involving saving: Outcome Check

Outcome check for Simple interest and problems involving saving: Work out simple interest and solve problems involving saving.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simple interest and problems involving saving: Outcome Check

Outcome check for Simple interest and problems involving saving: Define different terms such as simple interest, rate, principle, and time.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent expressions and number sequences: Lesson Opener

Learning focus: Equivalent expressions and number sequences

Country link: a Rwandan school, market, farming, building, transport, or data problem for Equivalent expressions and number sequences.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- write sequences of whole numbers, fractions and decimals
- give examples of algebraic expressions and their equivalent expressions
- explain how to find the rule for determining the n th term in a linear sequence
- match equivalent expressions
- determine the rule for a sequence
- find the missing number in a sequence following a particular rule

Inquiry questions:

- How can equivalent expressions and number sequences help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent expressions and number sequences: Learn

Equivalent expressions and number sequences teaches how numbers change in a regular way.

Core idea:

- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:

- Equivalent expressions and number sequences
- Equivalent
- expressions
- number
- sequences
- method
- model
- unit

Equivalent expressions and number sequences: Worked Example

Worked example for Equivalent expressions and number sequences: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: $12 - 6 = 6$, $18 - 12 = 6$.
2. Rule: add 6 each time.
3. $24 + 6 = 30$, $30 + 6 = 36$.

Answer: 30, 36.

Equivalent expressions and number sequences: Vocabulary And Study Skill

Use these words and study moves before you practise Equivalent expressions and number sequences. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Equivalent expressions and number sequences
- Equivalent
- expressions
- number
- sequences
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Equivalent expressions and number sequences without a clear example, method, or evidence.

Equivalent expressions and number sequences: Guided Activity

Guided activity for Equivalent expressions and number sequences:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Equivalent expressions and number sequences: Practice And Correction

Practice for Equivalent expressions and number sequences:

Main outcome: write sequences of whole numbers, fractions and decimals.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:

Answers: 25, 30. Rule: add 5 each time.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to equivalent expressions and number sequences and solve it.

Equivalent expressions and number sequences: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Equivalent expressions and number sequences.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent expressions and number sequences: Outcome Check

Outcome check for Equivalent expressions and number sequences: Write sequences of whole numbers, fractions and decimals.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent expressions and number sequences: Outcome Check

Outcome check for Equivalent expressions and number sequences: Give examples of algebraic expressions and their equivalent expressions.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Regular polygons and bearings: Lesson Opener

Learning focus: Regular polygons and bearings

Country link: a Rwandan school, market, farming, building, transport, or data problem for Regular polygons and bearings.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use triangle, angle properties and compass directions to solve real-world problems involving regular polygons and bearings
- define a regular polygon
- give the formulae used to calculate the perimeter and area of a regular polygon
- explain directions and bearing using compass points
- understand and use the angle sum of a triangle to determine the angle sum of a polygon
- derive the interior angle of a regular polygon

Inquiry questions:

- How can regular polygons and bearings help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Regular polygons and bearings: Learn

Regular polygons and bearings teaches how much flat surface is covered.

Core idea:

- Area is counted in square units.
- Rectangles can be found by length x width.
- Land and floor measurements need clear units and a labelled diagram.

Key words:

- Regular polygons and bearings
- Regular
- polygons
- bearings
- method
- model
- unit
- answer

Regular polygons and bearings: Worked Example

Worked example for Regular polygons and bearings: A rectangular garden is 9 m long and 4 m wide. Find its area.

1. Draw a rectangle and label length 9 m and width 4 m.
2. Area of a rectangle = length x width.
3. $9 \times 4 = 36$.

Answer: 36 square metres.

Regular polygons and bearings: Vocabulary And Study Skill

Use these words and study moves before you practise Regular polygons and bearings. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Regular polygons and bearings
- Regular
- polygons
- bearings
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Regular polygons and bearings without a clear example, method, or evidence.

Regular polygons and bearings: Guided Activity

Guided activity for Regular polygons and bearings:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Regular polygons and bearings: Practice And Correction

Practice for Regular polygons and bearings:

Main outcome: use triangle, angle properties and compass directions to solve real-world problems involving regular polygons and bearings.

1. Draw a rectangle that is 6 units long and 3 units wide.
2. Find its area in square units.
3. Explain why area uses square units.

Answer check:

Answer: a 6 by 3 rectangle has area 18 square units.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to regular polygons and bearings and solve it.

Regular polygons and bearings: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Regular polygons and bearings.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Regular polygons and bearings: Outcome Check

Outcome check for Regular polygons and bearings: Use triangle, angle properties and compass directions to solve real-world problems involving regular polygons and bearings.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Regular polygons and bearings: Outcome Check

Outcome check for Regular polygons and bearings: Define a regular polygon.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Construction of polygons and nets for cuboids and prisms: Lesson Opener

Learning focus: Construction of polygons and nets for cuboids and prisms

Country link: a Rwandan school, market, farming, building, transport, or data problem for Construction of polygons and nets for cuboids and prisms.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- construct polygons and design nets to make cuboids and prisms
- show how to construct polygons with given properties using a protractor, a ruler and, a pair of compasses
- demonstrate how a 2D shape can be folded to make a 3D solid and name the 2D shape used
- show that the net of a solid is not unique
- construct polygons using a protractor, a ruler, and a pair of compasses
- explore different strategies for constructing polygons with given properties

Inquiry questions:

- How can construction of polygons and nets for cuboids and prisms help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Construction of polygons and nets for cuboids and prisms: Learn

Construction of polygons and nets for cuboids and prisms teaches properties of three-dimensional shapes such as cubes, cuboids, and cylinders.

Core idea:

- Solid shapes have faces, edges, and vertices.
- A cuboid has rectangular faces.
- A cube has six equal square faces.
- A cylinder has two circular faces and one curved surface.
- Nets show how flat faces fold to make a solid.

Key words:

- Construction of polygons and nets for cuboids and prisms
- Construction
- polygons
- nets
- cuboids
- prisms
- method
- model

Construction of polygons and nets for cuboids and prisms:

Worked Example

Worked example for Construction of polygons and nets for cuboids and prisms: Name the faces of a cuboid.

1. Look at the solid shape.
2. A cuboid has 6 faces.
3. Each face is a rectangle.
4. Opposite faces are equal in size.

Answer: A cuboid has 6 rectangular faces, 12 edges, and 8 vertices.

Construction of polygons and nets for cuboids and prisms:

Vocabulary And Study Skill

Use these words and study moves before you practise Construction of polygons and nets for cuboids and prisms. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Construction of polygons and nets for cuboids and prisms
- Construction
- polygons
- nets
- cuboids
- prisms
- method
- model

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Construction of polygons and nets for cuboids and prisms without a clear example, method, or evidence.

Construction of polygons and nets for cuboids and prisms:

Guided Activity

Guided activity for Construction of polygons and nets for cuboids and prisms:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Construction of polygons and nets for cuboids and prisms: Practice And Correction

Practice for Construction of polygons and nets for cuboids and prisms:

Main outcome: construct polygons and design nets to make cuboids and prisms.

1. Draw or model a cube, cuboid, and cylinder.
2. Count faces, edges, and vertices where they apply.
3. Sketch a simple net for a cube or cuboid.

Answer check:

Answer check: cube has 6 equal square faces; cuboid has 6 rectangular faces; cylinder has circular faces and a curved surface.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to construction of polygons and nets for cuboids and prisms and solve it.

Construction of polygons and nets for cuboids and prisms: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Construction of polygons and nets for cuboids and prisms.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Construction of polygons and nets for cuboids and prisms: Outcome Check

Outcome check for Construction of polygons and nets for cuboids and prisms: Construct polygons and design nets to make cuboids and prisms.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Construction of polygons and nets for cuboids and prisms: Outcome Check

Outcome check for Construction of polygons and nets for cuboids and prisms: Show how to construct polygons with given properties using a protractor, a ruler and, a pair of compasses.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Lesson Opener

Learning focus: Area bounded by a circle, surface area of cuboids and volume of a cylinder

Country link: a Rwandan school, market, farming, building, transport, or data problem for Area bounded by a circle, surface area of cuboids and volume of a cylinder.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- calculate the area enclosed by a circle, the surface area of cuboids and the volume of a cylinder
- state the formula for finding the area enclosed by a circle and explain how it can be derived from the circumference of a circle
- explain the surface area of a cuboid as the sum areas of its net
- state the formula for finding the volume of a given solid shape
- calculate the area bounded by a circle
- use the net of a solid to determine its surface area

Inquiry questions:

- How can area bounded by a circle, surface area of cuboids and volume of a cylinder help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Learn

Area bounded by a circle, surface area of cuboids and volume of a cylinder teaches properties of three-dimensional shapes such as cubes, cuboids, and cylinders.

Core idea:

- Solid shapes have faces, edges, and vertices.
- A cuboid has rectangular faces.
- A cube has six equal square faces.
- A cylinder has two circular faces and one curved surface.
- Nets show how flat faces fold to make a solid.

Key words:

- Area bounded by a circle, surface area of cuboids and volume of a cylinder
- Area
- bounded
- circle
- surface
- area
- cuboids
- volume

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Worked Example

Worked example for Area bounded by a circle, surface area of cuboids and volume of a cylinder:
Name the faces of a cuboid.

1. Look at the solid shape.
2. A cuboid has 6 faces.
3. Each face is a rectangle.
4. Opposite faces are equal in size.

Answer: A cuboid has 6 rectangular faces, 12 edges, and 8 vertices.

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Vocabulary And Study Skill

Use these words and study moves before you practise Area bounded by a circle, surface area of cuboids and volume of a cylinder. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Area bounded by a circle, surface area of cuboids and volume of a cylinder
- Area
- bounded
- circle
- surface
- area
- cuboids
- volume

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Area bounded by a circle, surface area of cuboids and volume of a cylinder without a clear example, method, or evidence.

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Guided Activity

Guided activity for Area bounded by a circle, surface area of cuboids and volume of a cylinder:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Practice And Correction

Practice for Area bounded by a circle, surface area of cuboids and volume of a cylinder:

Main outcome: calculate the area enclosed by a circle, the surface area of cuboids and the volume of a cylinder.

1. Draw or model a cube, cuboid, and cylinder.
2. Count faces, edges, and vertices where they apply.
3. Sketch a simple net for a cube or cuboid.

Answer check:

Answer check: cube has 6 equal square faces; cuboid has 6 rectangular faces; cylinder has circular faces and a curved surface.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to area bounded by a circle, surface area of cuboids and volume of a cylinder and solve it.

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Area bounded by a circle, surface area of cuboids and volume of a cylinder.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Outcome Check

Outcome check for Area bounded by a circle, surface area of cuboids and volume of a cylinder: Calculate the area enclosed by a circle, the surface area of cuboids and the volume of a cylinder.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area bounded by a circle, surface area of cuboids and volume of a cylinder: Outcome Check

Outcome check for Area bounded by a circle, surface area of cuboids and volume of a cylinder: State the formula for finding the area enclosed by a circle and explain how it can be derived from the circumference of a circle.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Lesson Opener

Learning focus: Data handling

Country link: a Rwandan school, market, farming, building, transport, or data problem for Data handling.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- collect, represent and interpret data in order to solve a problem
- explain when it is appropriate to use a tally and how to obtain frequency from the tally
- explain how to use pie charts to represent proportions
- interpret line graphs as representation of data
- devise a question or hypothesis that requires data for its resolution
- decide what data to collect to answer a question

Inquiry questions:

- How can data handling help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Learn

Data handling teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Data handling
- Data
- handling
- method
- model
- unit
- answer
- check

Data handling: Worked Example

Worked example for Data handling: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Data handling: Vocabulary And Study Skill

Use these words and study moves before you practise Data handling. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Data handling
- Data
- handling
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Data handling without a clear example, method, or evidence.

Data handling: Guided Activity

Guided activity for Data handling:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Data handling: Practice And Correction

Practice for Data handling:

Main outcome: collect, represent and interpret data in order to solve a problem.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to data handling and solve it.

Data handling: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Data handling.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Outcome Check

Outcome check for Data handling: Collect, represent and interpret data in order to solve a problem.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Outcome Check

Outcome check for Data handling: Explain when it is appropriate to use a tally and how to obtain frequency from the tally.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing and comparing whole numbers beyond 1: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Reading, writing and comparing whole numbers beyond 1 that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Multiplication and division of integers: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Multiplication and division of integers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Powers, indices, LCM and GCF: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Powers, indices, LCM and GCF that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Operations on fractions: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Operations on fractions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Reading, writing and comparing whole numbers beyond 1: Explain this term using an example from the book, your school, or your community.
- Multiplication and division of integers: Explain this term using an example from the book, your school, or your community.
- Powers, indices, LCM and GCF: Explain this term using an example from the book, your school, or your community.
- Operations on fractions: Explain this term using an example from the book, your school, or your community.
- Rounding and conversion of decimals fractions/numbers: Explain this term using an example from the book, your school, or your community.
- Percentages and ratios: Explain this term using an example from the book, your school, or your community.
- Relationship between volume, capacity and mass: Explain this term using an example from the book, your school, or your community.
- Speed, distance and time: Explain this term using an example from the book, your school, or your community.
- Simple interest and problems involving saving: Explain this term using an example from the book, your school, or your community.
- Equivalent expressions and number sequences: Explain this term using an example from the book, your school, or your community.
- Regular polygons and bearings: Explain this term using an example from the book, your school, or your community.
- Construction of polygons and nets for cuboids and prisms: Explain this term using an example from the book, your school, or your community.
- Area bounded by a circle, surface area of cuboids and volume of a cylinder: Explain this term using an example from the book, your school, or your community.
- Data handling: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.